



Lesson 7.6 — Amplitude, Period, and Phase Shift

Big idea: Every sinusoid can be written $y = a \sin(b(x - h)) + k$. Each parameter does one job: a stretches vertically, b changes period, h shifts horizontally, k shifts vertically.

Quick review

Amplitude: $|a|$. Maximum displacement from the midline.

Period: $2\pi/|b|$. The horizontal length of one full wave.

Phase shift: h . Horizontal shift (if h positive).

Vertical shift / midline: k . The horizontal line $y = k$ that the wave oscillates around.

Reflection: if $a < 0$, the wave is flipped vertically.

Worked example

Worked example. Identify amplitude, period, phase shift, midline of $y = 3 \sin(2(x - \pi/4)) + 1$.

$a = 3 \rightarrow$ amplitude 3. $b = 2 \rightarrow$ period $2\pi/2 = \pi$. $h = \pi/4 \rightarrow$ shift right $\pi/4$. $k = 1 \rightarrow$ midline $y = 1$.

Warm-ups

1. State the amplitude and period of $y = 4 \sin x$.
2. State the period of $y = \sin(3x)$.
3. State the midline of $y = \sin x + 5$.

Practice

1. State amplitude, period, phase shift, and midline of $y = 2 \cos(x - \pi/2) - 3$.
2. State amplitude, period, phase shift, and midline of $y = (1/2) \sin(\pi x)$.
3. Write a sinusoidal equation with amplitude 5, period 4, midline $y = 0$, and no phase shift, using sine.

Challenge

1. Write the equation of a sine function with maximum value 7, minimum value -1 , period π , and a maximum at $x = 0$.
2. Write $y = 3 \cos(2x - \pi)$ in the form $a \sin(b(x - h)) + k$. (Hint: $\cos = \sin$ shifted by $\pi/2$.)



Answer key — Lesson 7.6

Check your work. If a problem tripped you up, rewatch the step in the video where that concept appears.

Warm-ups

1. Amplitude 4, period 2π .
2. Period $2\pi/3$.
3. $y = 5$.

Practice

1. Amplitude 2, period 2π , phase shift right $\pi/2$, midline $y = -3$.
2. Amplitude $1/2$, period $2\pi/\pi = 2$, no phase shift, midline $y = 0$.
3. $y = 5 \sin(\pi x/2)$. (Period $2\pi/(\pi/2) = 4$ ✓.)

Challenge

1. Midline $= (7 + (-1))/2 = 3$. Amplitude $= (7 - (-1))/2 = 4$. Period $\pi \rightarrow b = 2\pi/\pi = 2$. Maximum at $x = 0$ suggests cosine: $y = 4 \cos(2x) + 3$.
2. $y = 3 \cos(2(x - \pi/2)) = 3 \sin(2(x - \pi/2) + \pi/2) = 3 \sin(2x - \pi + \pi/2) = 3 \sin(2(x - \pi/4))$. So the equivalent sine form is $y = 3 \sin(2(x - \pi/4))$.